

Galois group of $(x^2-2)(x^2-3)$ over $\mathbb{Q}$.

Since the polynomial $(x^2-2)(x^2-3)$ has four distinct roots $\pm\sqrt{2}, \pm\sqrt{3}$, it is separable.

And, since $\sqrt{3} \notin \mathbb{Q}(\sqrt{2})$, $[\mathbb{Q}(\sqrt{2}, \sqrt{3}) : \mathbb{Q}] = [\mathbb{Q}(\sqrt{2}, \sqrt{3}) : \mathbb{Q}(\sqrt{2})] \cdot [\mathbb{Q}(\sqrt{2}) : \mathbb{Q}] = 4$.

So, $|\text{Aut}(K/\mathbb{Q})| = [K : \mathbb{Q}] = 4$ where $K = \mathbb{Q}(\sqrt{2}, \sqrt{3})$, the splitting field over $\mathbb{Q}$.

Since the automorphism leaving $\mathbb{Q}$ fixed must be conjugate the roots,

there are four possibilities:

$$\begin{cases} \sqrt{2} \mapsto \sqrt{2} \\ \sqrt{3} \mapsto \sqrt{3} \end{cases}, \begin{cases} \sqrt{2} \mapsto -\sqrt{2} \\ \sqrt{3} \mapsto \sqrt{3} \end{cases}, \begin{cases} \sqrt{2} \mapsto \sqrt{2} \\ \sqrt{3} \mapsto -\sqrt{3} \end{cases}, \begin{cases} \sqrt{2} \mapsto -\sqrt{2} \\ \sqrt{3} \mapsto -\sqrt{3} \end{cases}.$$

Let $\sigma : K \rightarrow K; a + b\sqrt{2} + c\sqrt{3} + d\sqrt{6} \mapsto a - b\sqrt{2} + c\sqrt{3} - d\sqrt{6}$

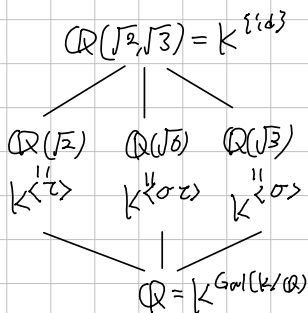
$\tau : K \rightarrow K; a + b\sqrt{2} - c\sqrt{3} - d\sqrt{6} \mapsto a + b\sqrt{2} - c\sqrt{3} - d\sqrt{6}$.

(That is, $\sigma : \begin{cases} \sqrt{2} \mapsto -\sqrt{2} \\ \sqrt{3} \mapsto \sqrt{3} \end{cases}$ and $\tau : \begin{cases} \sqrt{2} \mapsto \sqrt{2} \\ \sqrt{3} \mapsto -\sqrt{3} \end{cases}$, and we can observe that

$$\sigma\tau : \begin{cases} \sqrt{2} \mapsto -\sqrt{2} \\ \sqrt{3} \mapsto -\sqrt{3} \end{cases}.$$

Now, $\text{Gal}(K/\mathbb{Q}) = \{1, \sigma, \tau, \sigma\tau\} \cong \mathbb{Z}_2 \times \mathbb{Z}_2$, corresponding σ to $(1, 2)$ and τ to $(2, 1)$.

Moreover,



Galois group of $x^3 - 2$ over $\mathbb{Q}$.

Denote $\zeta_3 = e^{i\frac{2\pi}{3}} = \frac{1}{2}(-1 + i\sqrt{3})$, the 3th primitive root of unity.

Since $\zeta_3^3 = 1$, the set $\{\sqrt[3]{2}, \sqrt[3]{2}\zeta_3, \sqrt[3]{2}\zeta_3^2\}$ are all roots of $x^3 - 2$ over $\mathbb{Q}$.
So, the splitting field is $K = \mathbb{Q}(\sqrt[3]{2}, \zeta_3) = \mathbb{Q}(\sqrt[3]{2}, \sqrt{3})$.

Observe that: the minimal polynomial of $\sqrt[3]{2}$ over $\mathbb{Q}$ is $x^3 - 2$, and
" ζ_3 over $\mathbb{Q}$ is $x^2 + x + 1$.

And, since $x^2 + x + 1$ is irreducible over $\mathbb{Q}(\sqrt[3]{2})$, (it cannot have root in $\mathbb{Q}(\sqrt[3]{2})$)

$$[\mathbb{Q}(\sqrt[3]{2}, \zeta_3) : \mathbb{Q}] = [\mathbb{Q}(\sqrt[3]{2}, \zeta_3) : \mathbb{Q}(\sqrt[3]{2})] \cdot [\mathbb{Q}(\sqrt[3]{2}) : \mathbb{Q}] = 2 \cdot 3 = 6.$$

the tower lemma gave. And, since $x^3 - 2$ is separable, $|\text{Aut}(K/\mathbb{Q})| = [K : \mathbb{Q}] = 6$.

Now, construct explicitly the $\text{Aut}(K/\mathbb{Q})$.

Since any automorphism of K leaving $\mathbb{Q}$ fixed must conjugate the roots,

it must map $\sqrt[3]{2}$ to $\sqrt[3]{2}$ or $\sqrt[3]{2}\zeta_3$ or $\sqrt[3]{2}\zeta_3^2$

and maps ζ_3 to ζ_3 or ζ_3^2 .

So, there are exactly 6 possibilities. Consider two automorphisms

σ satisfying $\sigma(\sqrt[3]{2}) = \sqrt[3]{2}\zeta_3$ and $\sigma(\zeta_3) = \zeta_3$

and τ satisfying $\tau(\sqrt[3]{2}) = \sqrt[3]{2}$ and $\tau(\zeta_3) = \zeta_3^2 = -1 - \zeta_3$.

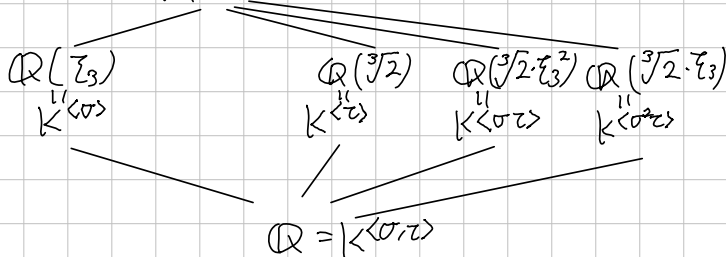
Then, $\sigma^3: \begin{cases} \sqrt[3]{2} \mapsto \sqrt[3]{2}\zeta_3 \mapsto \sqrt[3]{2}\zeta_3^2 \mapsto \sqrt[3]{2}\zeta_3^3 = \sqrt[3]{2} \\ \zeta_3 \mapsto \zeta_3 \end{cases}$, $\tau^2: \begin{cases} \sqrt[3]{2} \mapsto \sqrt[3]{2} \\ \zeta_3 \mapsto \zeta_3^2 \mapsto \zeta_3^4 = 1 - \zeta_3 = \zeta_3 \end{cases}$

are actually identities. Moreover,

$\sigma^2: \begin{cases} \sqrt[3]{2} \mapsto \sqrt[3]{2}\zeta_3^2 \\ \zeta_3 \mapsto \zeta_3 \end{cases}$, $\sigma\tau: \begin{cases} \sqrt[3]{2} \mapsto \sqrt[3]{2}\zeta_3 \\ \zeta_3 \mapsto \zeta_3^2 \end{cases}$, $\sigma^2\tau: \begin{cases} \sqrt[3]{2} \mapsto \sqrt[3]{2}\zeta_3^2 \\ \zeta_3 \mapsto \zeta_3^2 \end{cases}$, $\tau\sigma: \begin{cases} \sqrt[3]{2} \mapsto \sqrt[3]{2}\zeta_3^2 \\ \zeta_3 \mapsto \zeta_3 \end{cases}$

So, $\text{Aut}(K/\mathbb{Q}) = \{1, \sigma, \sigma^2, \tau, \sigma\tau, \sigma^2\tau\} \cong S_3$, corresponding σ to $(1\ 2\ 3)$, τ to $(1\ 2)$

Moreover, $\mathbb{Q}(\sqrt[3]{2}, \zeta_3) = K^{\langle \text{id} \rangle}$



How to find a fixed field corresponding to $\langle \sigma\tau \rangle$?

As we know, the $\text{Tr}_{K/K^{\langle \sigma\tau \rangle}}$ is invariant under any automorphism in $\langle \sigma\tau \rangle$.

So, Calculate:

$$\text{Tr}(\sqrt[3]{2}) = \text{id}(\sqrt[3]{2}) + (\sigma\tau)(\sqrt[3]{2}) = \sqrt[3]{2} + \sqrt[3]{2}\zeta_3 = \sqrt[3]{2}(1 + \zeta_3) = \sqrt[3]{2} \cdot (-\zeta_3^2).$$

Therefore, $\sqrt[3]{2}\zeta_3^2$ is fixed by id and $\sigma\tau$, so $\mathbb{Q}(\sqrt[3]{2}\zeta_3^2) \subseteq K^{\langle \sigma\tau \rangle}$.

Moreover, $[\mathbb{Q}(\sqrt[3]{2}, \zeta_3) : \mathbb{Q}(\sqrt[3]{2}\zeta_3^2)] = 2$, so the Galois theorem gives the result.

Galois group of $x^8 - 2$ over $\mathbb{Q}$.

Galois group of $x^5 - 3$ over $\mathbb{Q}(\zeta)$, where ζ is a primitive fifth root of unity. Observe that: $x^5 - 3$ has five distinct roots, $\sqrt[5]{3} \cdot \zeta^k$ ($k=0,1,\dots,4$).

Since the minimal polynomial of ζ over $\mathbb{Q}$ is $x^4 + x^3 + x^2 + x + 1$, therefore $[\mathbb{Q}(\zeta) : \mathbb{Q}] = 4$. And, the minimal polynomial of $\sqrt[5]{3}$ over $\mathbb{Q}$ is $x^5 - 3$, therefore $[\mathbb{Q}(\sqrt[5]{3}) : \mathbb{Q}] = 5$. Therefore, the splitting field of $x^5 - 3$ over $\mathbb{Q}$,

$$K = \mathbb{Q}(\sqrt[5]{3}, \sqrt[5]{3}\zeta, \sqrt[5]{3}\zeta^2, \sqrt[5]{3}\zeta^3, \sqrt[5]{3}\zeta^4) = \mathbb{Q}(\sqrt[5]{3}, \zeta) = \mathbb{Q}(\sqrt[5]{3})\mathbb{Q}(\zeta)$$

has degree of 20 over $\mathbb{Q}$, that is, $[K : \mathbb{Q}] = 20$.

More precisely, by the tower lemma $[K : \mathbb{Q}]$ is divided by $[\mathbb{Q}(\sqrt[5]{3}) : \mathbb{Q}]$ and $[\mathbb{Q}(\zeta) : \mathbb{Q}]$.

Since 5 and 4 are relatively prime, $[K : \mathbb{Q}] \geq 20$.

And, $x^5 - 3 \in \mathbb{Q}(\zeta)[x]$ has $\sqrt[5]{3}$ as a root, so

$$[\mathbb{Q}(\sqrt[5]{3}, \zeta) : \mathbb{Q}(\zeta)] \cdot [\mathbb{Q}(\zeta) : \mathbb{Q}] \leq 20.$$

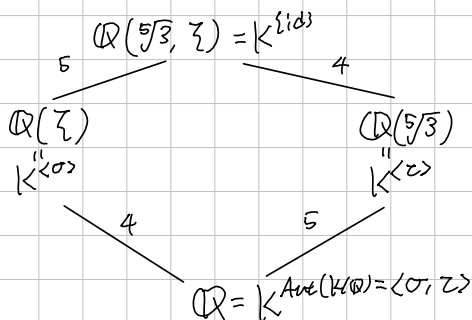
Consequently, $[K : \mathbb{Q}] = 20$. Now, by the Fundamental Theorem of Galois Theory, $K/\mathbb{Q}(\zeta)$ is also Galois, with $[K : \mathbb{Q}(\zeta)] = 5$.

Therefore, $\text{Gal}(K/\mathbb{Q}(\zeta))$ has order of 5, so it must be a cyclic group of order 5. Consequently, the automorphism

$$\sigma : \begin{cases} \sqrt[5]{3} \mapsto \sqrt[5]{3} \cdot \zeta \\ \zeta \mapsto \zeta \end{cases} \quad (\text{It is given by the conjugate isomorphism theorem}).$$

has order of 5, therefore $\text{Gal}(K/\mathbb{Q}(\zeta)) = \langle \sigma \rangle \cong \mathbb{Z}_5$. $\square$

Latticisally,



where $\tau : \begin{cases} \sqrt[5]{3} \mapsto \sqrt[5]{3} \\ \zeta \mapsto \zeta^2 \end{cases}$ Therefore, $\text{Aut}(K/\mathbb{Q}) \cong \langle (1\ 2\ 3\ 4\ 5), (2\ 3\ 5\ 4) \rangle$.

Galois group of x^6+3 over $F = \mathbb{Q}$ or $\mathbb{F}_5$ or $\mathbb{F}_7$. (UCLA QQual, 2022 Fall)
For each case, determine the splitting field K and evaluate $[K:F]$.
And, find the Galois group $\text{Gal}(K/F)$.
(Note: the field of characteristic 0 or finite field are perfect.)

